# Introduction to Psychology and its methods AI

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# Week 1

## Lecture 1

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Multiple choice exam (90%) + quizzes (10%)
ø
What is psychology?

- The scientific study of mind, brain and behavior/ of mind and brain to explain behavior

Psychology uses scientific methods to explain social concepts, because common sense fails

Forer effect: the feeling that something generic fits you specifically

Topic 1: Genes and Evolution
Genetic basis

- A violent person has violent genes?
    - 70s: no → environment leads to violent behavior
    - 90s: yes
    - Now: ? both genes and environment is important
- Bio-psycho-social model (all are important for behavior)
- Source of Similarities and differences
- The genetic bias:
    - Cell → nucleus → 46 chromosomes → DNA → genes (govern cells functioning; get translated into proteins)
    - Genes can be turned on or off (if it is turned off, it does not produce that protein)/ gene expression
        - Active genes = expressed genes
        - E.g. a skin cell has different genes activated than a tooth cell
        - Biochemical environment inside the cell controls the gene expression
            - Biochemical environment inside cell is influenced by biochemical environment outside of it, timing in development, overall environment, behavior…
    - Genome
        - Contains set of complete instructions for making an organism
        - Contains all genes in that organism
    - Molecular behavior genetics
        - Heredity influences you (e.g. bodyweight), but there is no single gene for a something (e.g. no single fat gene, but a combination of different genes)
    - Genotype → specific set of genes for one individual organism
    - Phenotype → the set of observable characteristics or traits of an organism (can be changed by environment, so genotype + environment basically)
    - Monozygotic (identical) twins have same genes/ dizygotic (fraternal) twins
    - Study of nature or nurture:
        - Twins, siblings, adopted children are all studied
        - If identical twins are more similar than fraternal ones that hints on genetics being important for behavior
    - Each gene is paired with another gene
        - The pairs are located at corresponding positions of pairs of chromosomes
    - Allele: one specific variant of a gene
        - Dominant: you only need one gene to get that characteristic (e.g. 1 brown-eye-color gene (which is dominant), and one green eye-color gene -→ brown eyes)
        - Recessive: both of the genes need to be the same to have that characteristic (2 green eye color genes)

- Polygenetic inheritance (you need multiple genes for the trait)
- Some human traits are fixed (especially biological feature)
- Most psychological traits are not fixed
- Darwin:
  - Genome is shaped by evolution over the years → Darwin's evolution theory
    - All organisms descendants from small set of ancestors
- Natural selection
  - Three conditions
    - Variation among individuals of a population (variation makes sure that genes survive)
    - Individuals with a certain trait survive and reproduce at higher rate
    - The trait associated with this advantage will be passed from parent to offspring and will be better represented in the next generation
  - Organisms differ in genotype and variations in genotype are passed from generation to generation
  - It is the survival of the gene not the individual that matters
  - Evidences for modern evolutionary theory:
    - Fossil records
    - Resemblances between genomes of various organisms
    - Pseudogenes (lacks regulatory elements → are not just inactive but not functional)
      - E.g. the gene that produces the enzyme glow → produces vitamin C → we have that only as a pseudogene
    - Distribution of species across the world (Pangaea etc.)
      - Oceanic islands (made by volcanic activity → typical land animals don't exist there, only air and see animals can get and evolve there)
  - Anthropoid apes and humans share many genes → but the apes have one more set of chromosomes (24 pairs in total), for us two chromosomes merged (23 pairs in total)
  - Despite overwhelming evidence, many people remain skeptical about the theory of evolution
  - Things evolution doesn't (necessarily) do:
    - Improve organisms

Experience and brain development:

- Early brain development important for total brain development and usage
  - Rat experiment → rat that was stripped of everything but food and water developed less thick cortex than other rats
  - Almost irreversible in future
- Human contains most of its neurons it will ever have from birth
- Connections amongst neurons get build be experience
- Neurons: Use it or lose it

- MAOA gene → people with low level of MAO are expected to be more violent → BUT environment also plays a huge role → nature and nurture work together
- Cultural influences
    - Behavior cannot be understood outside their contexts
    - Norms (rules for accepted and expected behavior)
    - Individualistic and collectivistic cultures (but varies within)

## Lecture 1.2

The Brain and the Nervous System

Studying the brain

- Phrenology
    - 19th century pseudo-science
    - Bumps on skull were interpreted in terms of personality traits
    - Used as 'proof' for racist ideas
    - Scientists discovered structures and areas of the brain in proving phrenology wrong
    - Phineas Gage (dude who got temping iron shot through his skull)
        - Lived, but personality changed after incident
        - Destroyed much of his prefrontal cortex
            - Involved in impulse control
        - Survived for 10 years
- Methods of studying the brain
    - Clinical observations of patients with brain demage
    - Brain scans
    - Experimental techniques
        - Invasive animal studies
            - Simple cells recordings
        - TMS (Transcranial magnetic stimulation) → shortly disrupts region of brain so you can see what it is involved in
        - EEG (electroencephalogram) → shows electrical activity of the brain
            - ERP → Event related potentials → always have a specific event you are interested in, like showing a dog several times
        - Brain imaging
            - fMRI (magnetic resonance imaging, but also gets functional data like blood float and brain activity (only average over 3 sec))
            - pet scans (more invasive)
            -

Building blocks of the nervous system

- 2 kinds of basic cells
    - Glia cells
        - Protectors, non-neuronal, no electrical impulses
    - Neurons
        - Produce electrical impulses
        - Different types
            - Sensory receptors (temperature etc.)
            - Sensory (afferent) neurons

            - Motor
            - Add rest
        - Add picture of neuron

Communication among neurons

        - Communication among neurons
            - Action potential:
            - Very fast
            - Add stuff
        - Neurons either fire or do not fire (all or none law)
            - Intensity variations by
                - Variation in the number of neuron fireing (number)
                - Variation in fireing rate (trequency)
        - Neurons interact
            - Via synapses
                - Place where a signal passes from one nerve cell to other
            - Through chemical substances/Neurotransmitters
                - Chemical substances that transmit signals from one neuron to another
                - Lock-and-key model
                - Effect is terminated by
                - Autoreceptors
                - Synaptic reuptake
                - Enzyme deactivation
                - The binding of a neurotransmitter with a receptor …. Add later
        - Add stuff about Drugs, Agonists and Antagonists
        -

Communication of the brain with the body

- How neurotransmitters influence us
    - Serotonin cycles/pathways → mood regulation
    - Dopamine pathways → schizophrenia, Parkinson disease, movement
- Add tree about nervous system

The brain

- Left hemisphere is in control of the right part of the body and right hemisphere of the left part
- Brainstem as the crossover point
- Cerebellum (motor functions)
- Hypothalamus: regulates body funtions and hormons
- Amygdala: where emotional information passes through for other parts of the brain to process it
- Major sections
    - Frontal lobe
    - Parietal lobe
    - Temporal lobe
    - Occipital lobe
- Visual cortex (in occipital lobe)
- Motor cortex (output/motion) and sensory cortex (input/sensing)

- Broca's area → impaired speaking
- Wernicke's area → impaired understanding
- Association areas:
    - ¾ of human cortex
    - Found in many animals
- Brain re-organization
    - Plasticity
        - Brains capacity for modification
        - The brain is sculpted by our genes by also by our experiences
- Our divided brain
    - 1960s left hemisphere was termed the dominant brain → not true
    - The left hemisphere processes reading, writing, speaking, mathematics, and comprehension skills, logic, complex motor behavior, language
    - Right hemisphere: spatial ability, emotion (facial expression, mood), musical ability (perception of melody), some memory tasks (learning tasks when context doesn't matter)
    - Corpus callosum

Summary:

# Week 2

## Lecture 2.1 - Consciousness

Consciousness is hard to define

Our conscious experience is very limited

History of consciousness

- 60s: mental concepts like consciousness started reentering science of psycology (before seen as not scientific)

Introductions

- Definition
    - Subjected experiences of the world, resulting from brain activity
        - Brain and mind are inseparable
        - Subjective experience
            - We cannot know if two people experience something the same way (explanation gap) → Qualia
    - Subjective awareness of ourselves and our environment
    - An awareness of sensations, thoughts, feelings one is attending at given moment
    - More controlled/ focused when learning a complex concept or behavior
    - Becomes automatic by practice
- States of consciousness
    - Spontaneous: (Day)dreaming
    - Physiologically induced: Hallucination, food/oxygen starvation
    - Psychologically induced: sensory deprivation, meditation
- There is no specific region of the brain for consciousness BUT specific types of stimuli are located in specific brain regions

- Spatial neglect: when people ignore one side of space, sometimes not even aware that they are missing something, only realize something is wrong when its repeatedly pointed out to them (always not aware of left side) → eye etc. is functional, but the part of the region that processes it isn't functional, also awareness problem
- Specific patterns of brain activity can predict what a person is seeing
    - Specific region that recognizes faces
    - Specific separate region that recognized places
    - We can predict what a person is seeing based on their brain activity
- Cognitive neuroscience
    - Responsiveness in comatose patients
        - 5-10% have some degree of conscience
- Somatoparaphrenia
- After the fact explanations:
    - Unconscious acts that the mind struggles to explain → the conscious brain gives random explanation afterwards
    - Free will question
    - Also split brain experiment again
        - + Left hemisphere comes up with random explanations for actions decided by right hemisphere
- Telling more than we can know (similar to after-the-fact-explanations)
- Readiness potential
    - Readiness potential: -535 msec
    - Awareness of decision: -204 msec
    - Action: 0 msec
    - → consciousness awareness comes after the brain decides to move the arm
    - Question of free will

Variation in normal consciousness

Attention

- Definition: Process that enables us to focus selectively on some things and avoid focusing on others
- Rapid toggling between activities is bad for attention
- Dichotic listening (shadowing): two things heard simultaneously, but even if you concentrate on one of the channels you would still recognize your name being said on the other (cocktail party phenomenon)
    - We follow meaning automatically
    - Unattended information may sometimes affect behavior
- Attention is about survival
- Subliminal perception: influencing your decision without you noticing it (like flashing coca cola image) → doesn't work, very minimal influence

Sleep

- Circadian cycle (about 24h)
    - Depends on light
        - Suprachiasmatic nucleus → pineal gland →
    - Childhood: most sleep → then decreases, but amount of non-REM sleep stays the same
    - 90min cycles for young adults → way faster for old adults
    - Sleep stages can be measured in brain waves (EEG)
        - Stages: REM, N1, N2, N3

        - REM sleep: brain shows same activity/waves as when awake, but you are asleep
- Why do we sleep? Multiple theories
    - Restorative theory
        - Protects us (we don't move when its dark and dangerous outside)
        - We get to rest
        - It supports growth
    - Circadian rhythm theory
        - Processing of memories etc.
    - Facilitates learning
- Dreams:
    - products of altered state of consciousness in which images and fantasies are confused with reality
        - Freud: unconscious desires lived out
        - Non-REM dreams: dull
        - REM dreams: bizarre and intense
    - During sleep: prefrontal cortex not that active, but visual areas etc. remain active

Drugs

- When is it a problem?
    - Tolerance:
        - Increasing amount of drug needed to achieve intended effect → BUT organs can't handle them better than before
    - Addiction
        - Chemical and physical, and psychological dependence
    - Withdrawal:
        - Physiological and psychological state characterized by feelings of anxiety, tension, and cravings from the addictive substances
- Alcohol is a depressant → first depresses inhibition and active control → the more amount the more functions get depressed (then you feel bad)
- Kinds of drugs
    - Simulants:
        - Increase behavioral and mental activity and activate the sympathetic nervous system
    - Depressants:
        - Reduce behavioral and mental activity by depressing the central nervous system
    - Opiates (narcotics):
        - Depress or slow down the central nervous system; relieve pain and suffering
    - Hallucinogens (psychedelics):
        - Produce alterations in cognition, mood and perception
    - Many commonly used drugs don't fit into those four major categories and overlap etc. (e.g. Marijuana)

## Lecture 2.2 – Sensation and perception: Vision

Perception is an active process

Sensation ≠ Perception

- Prosopagnosia example
- Human ears are most sensitive to sound frequencies that include human voices
- Our sensations gibe us information about what is out there, but only a small part of it (only certain frequencies etc.)
- Sensation: bottom up process
    - Detection of external stimuli (distal stimulus)
    - distal stimulus → proximal stimulus (add more)
- Perception: top down process
    - Add more ()
- We have restricted awareness:
    - Absolute threshold
        - The second you detect something it's crossed your absolute threshold
        - The threshold where you can detect a stimulus at least 50% of time
    - Difference threshold
        - When you can determine a difference in the stimulus at least 50% of the time (e.g. change in volume)
        - Weber's law: the more intense the stimulus, the bigger the difference needs to be

The senses

- Vision
- Hearing
- Taste
- Smell
- Skin senses (pressure, temperature, pain)
- Proprioception (you know where your body is also in relation to things around you)
- Interoception (hunger, internal organs feelings etc.)
- Vestibular sense
- Kinesthesia

- Quantitative variations (intensity)
    - Rate of neuron firing
    - Quantity/total number of neurons firing
- Qualitative variation (sensory quality)
    - Different neurons firing
    - Different patterns of neurons firing/pattern of activation
- Sensory adaptation
    - Tendency to respond less to a stimulus that has been present and unchanging for some time
    - Benefit: focus on informative changes in our environment

Vision

- Retina
    - Light (direct or reflected)
        - Can vary in intensity (great amplitude → brighter)
        - Can vary in wavelength
        - Light → cornea → Pupils, Iris → Lens → Retina → Optic nerve → brains visual cortex
        - Is absorbed by pigments and (add stuff)

        - Rods: night vision, b/w
        - Cones: day vision, colors
        - Two opposing theories:
            - Trichromatic Theory (Young-Helmholz)
                - Blue, red and green sensitive cones (small, medium and large wavelengths)
            - Opponent-process Theory (Hering)
                - There is no such thing as blue-yellow or red-green
                - Afterimages
                - There are opposing processes in your eye (black/white, blue/yellow, red/green)
            - Both are correct (neurons process it as opponent processing and 3 different types of cones like trichromatic theory)
        - Distribution of rods and cones across the retina
            - Fovea (where you are focusing on) has the most cones, further to the side are more rods
            -
- Beyond retina
    - Temporal lobe: information of what? → ventral pathways
    - Parietal lobe: Information of how/where? → dorsal pathway
- Perception occurs on the basis of
    - Bottom up data driven process
        - Starts at sensory receptors and works up to higher levels of processing
    - Top down knowledge driven perception
        - Our immediate context, motivation and emotion also affect our interpretation
        - Our experiences build us concepts and schemas that help organize and interpret (less ambiguity)
- Gestalt principles → perceptual interpretation on the basis of organizing principles
    - Closure, proximity etc.
- Perception of depth
    - Based on experiences since childhood
    - Depth cues
        - Binocular cues
            - Binocular Disparity
                - Cross disparity (how you see things double and 2 eyes things)
            - Conversion
                - Muscle information of how close your eyes are (their angle) sent to brain
        - Occlusion
        - Linear perspective
        - Shadow
        - Texture… add later
        - Motion parallax
            - Your own movement in a space (near thing move faster than far away ones)
- Perception of size

        - Depends on perceived distance
- Perceptual constancies:
    - We correctly perceive different properties of distal stimuli despite continuously changing proximal stimuli
        - E.g. Size, shape, lightness, color constancies
    -

# Week 3

## Lecture 3.1 – Learning

Non-associative learning

- Learning about a stimulus such as a sight or a sound
    - Habituation (become unaware)
        - Decrease in an innate response to a frequently repeated stimulus
            - Allows to focus on more important things
            - Relies on memory
        - Opposite: dishabituation → happens if what you were neglecting in the environment changes/ momentary change in something familiar (e.g. a sound suddenly stops)
    - Sensitization (become more aware)
        - Increase response to a stimulus after repeated exposure (e.g. fly flying around)
        - More generalized reactivity
        - Opposite: desensitization → extinction of sensitization (e.g. you get distracted by something else)

Associative learning

→ learning the relationship between things

- Classical conditioning
    - Form of learning where a neutral stimulus becomes significant for something e.g. digestive system → dogs digestive system started working when a bell rang, because the dog learned bell ringing = food
    - Before conditioning:
        - Unconditioned stimulus → you don't need conditioning ton respond to the stimulus (unconditioned response)
    - During conditioning
        - Neutral stimulus is introduces (e.g. certain sound) that illicit no response
        - Neutral stimulus is presented at the same time as unconditioned stimulus
    - After conditioning
        - Previously neutral stimulus becomes conditioned stimulus and evokes same (conditioned) response as the unconditioned response
    - Second order conditioning:
        - When conditioned stimulus (CS) is paired with new stimulus (S), the new S produces (conditioned response) CR
    - Extinction:
        - Association between CS and CR can be eliminated by repeatedly presenting the CS alone

- Spontaneous recovery
    - After extinction and a resting interval (connection appears again, but not as strong as before)
- Stimulus generalization
    - The CR can also be elicited by stimuli that are similar to the CS
- Discrimination
    - Only very similar CS will elicit CR (kinda opposite to stimulus generalization)
- What learned is expectancy
- Conditioning of emotions
    - noise → fear
    - noise + rat → fear
    - rat → fear
    - similar stimuli to rats → fear
- Limits
    - BUT an organism cannot learn EVERY association
        - Biological constraints → biological preparedness
    - Requirements of conditions
        - Contiguity (together in time) → learning occurs when CS is presented before US
        - Contingency – the CS has to be informative with respect to the change of the US is offered/ CS has to be reliable
            - 40% accuracy is enough
            - CS as a sort of predictor of the US
                - The CR is a reaction in preparation of the US
- Compensatory reactions of classical conditioning
    - Drugs (drug tolerance and cravings)

- Operant conditioning/instrumental conditioning
  → consequence of an action/response determines the likelihood that action will be repeated (punishment/reward)
    - About likelihood → positive response increases likelihood of repeated action, negative response decreases the likelihood of repeated action
    - Dr. Skinner → skinner boxes (experiments)
    - Shaping
        - Successive approximations
        - Teaching pigeons to fully turn around: Pigeon turns a little bit → reinforced → Pigeon turns more → reward → eventually pigeon turns around fully
        - Similar behaviors rewarded until full task is performed
    - (Watch pigeon religion)
    - Reinforcers:
        - Primary and secondary reinforcers
            - Primary: we naturally think its important (like food)
            - Secondary: we have learned that it is important (like money)
        - Positive and negative reinforcers (removal of unpleasant) (so not a punishment, but e.g. headache goes away after taking a painkiller)
        - More powerful than punishments
    - Punishment:

        - Positive and negative punishment (negative = removal of pleasant)
    - Continuous reinforcement
    - Partial reinforcement → schedules of reinforcement
        - Fixed-ratio (FR), e.g. food after clicking button exactly 4 times
        - Variable-ratio (VR), e.g. food after clicking button on average 4 times
        - Fixed-Interval (FI), e.g. food after 1st click after 4 minutes
        - Variable-Interval (VI), e.g. food after 1st click after on average4 minutes

      → partial reinforcement leads to less extinction than continuous reinforcement
    - Limits
        - Organisms cannot learn ANY association
            - Biological constraints → build-in predisposition
        - Requirements:
            - Contiguity
                - Learning only occurs when the reinforcer or punishment immediately follows the response
            - Contingency
                - Reinforcement or punishment should occur more likely after the response than otherwise
                - Without contingency → learned helplessness (dog and pain experiment)
- Humans have complex social lives that do not always correspond with biological constraints in obvious ways
- Turning play into work → rewards not always good (crayons and preschoolers/ reading and kids)

Observational learning/social learning
- learning also occurs in absence of responses
    - no need to show the behavior without reward
- we learn by observing others and imitating others
- vicarious learning (learning through other people's experiences)
- mirror neurons (look up again)
- Edward C Tolman → end of behaviorism

## Lecture 3.2 – Memory

**The study of memory**
- Memory:
    - Organization of the stimuli from the past
    - Nervous systems capacity to retain and bring back information
    - There are multiple memory systems with different rules
    - Encoding
    - Storage and consolidation
    - Retrieval
        - Recall
        - recognition

**Encoding**

- three memory systems: stage theory of memory
    - sensory input → sensory register → short term memory → rehersal → long term memory (in between always some degree of forgetting, first two stages can also be skipped)
- **sensory memory**:
    - feeds active working memory, recording momentary images, sounds, scents etc.
    - is fleeting, like a flashing light; can be perceived in entirety, but is forgotten very fast
    - 2 types
        - **Iconic memory**: picture/image that only lasts a few tenths of a second
        - **Echoic memory**: same with auditory stimuli, can be recalled within 3 or 4 seconds
- Short term memory/STM:
    - Limited capacity
        - Memory span task: magic number seven
            - You can keep about 7 items in your memory at a time
                - Item disappear as a result of decay and replacement (e.g. dementia)
            - By recording (chunks) → expansion of capacity
    - About half a minute time span
- Long term memory/LTM
    - Unlimited capacity
- Through rehearsal STM → LTM
    - Free recall task
    - Some suggestions against this stage theory
    - Instead: memory is an active process requiring attention → modern memory theory
- Modern Memory theory
    - STM → working memory
        - Information one is thinking about
    - LTM
        - Information, stored during longer interval, which is currently not actively used; deeper learning encodes memory
        - Depth of processing
            - Shallow processing (surface form)
            - Deep processing (meaning)
        - Understanding = memory booth
            - A schema may help to understand and remember
- Good encoding provides active means for later retrieval
- Effortful processing
    - Requires attention and conscious effort
- Automatic processing
    - Unconscious encoding of incidental information
    - Familiar or well learned information
- Visual encoding is a more powerful tool for encoding than audio etc.

Storage

- Memory consolidation

        - Neural process: transit fragile potential memory into more permanent state
        - Long-term potentiation (LTP)
            - Strengthening synaptic connection through repetition
            - More neurotransmitters and receptors build → higher probability of neuron firing
            - Can also disappear: long-term depression
            - Form of plasticity
        - Where are memories stored?
            - Everywhere, no hub of memory; pattern of activation across the brain
            - Information that passes through the hippocampus gets consolidated (left side more verbal, right side spatial memories)
            - Sleep supports memory consolidation
- The Amygdala, Emotions and Memory
    - Stress hormones can focus memory (+ boosts glucose production)
    - Flashbulb memories
        - Very clear, vivid memories of what you were doing or of how you found out about an event
            - Global event that is either extremely happy or extremely traumatic events (e.g. Moon landing, 9/11 etc.)
        - People think they remember those events accurately, but these memories are as susceptible to decay as every other memory
        - Reconsolidation can corrupt memory

Retrieval

- Recognition
    - Identify among choices (like multiple choice)
- Recall
    - Retrieve the information using effort (like fill in the blank)
- Good retrieval cues
    - Recreate context in which learning has occurred → context reinstatement
    - Reinstate the state of mind in which the original learning occurred
    - Richer encoding (context, understanding etc.) = richer recall
- Forgetting
    - Inability to retrieve information
    - Amnesia: deficit in long-term memory
        - Retrograde → loose old memories
        - Anterograde → can't form new memories
            - BUT implicit track still functional (so classical conditioning works) because it doesn't need to go through specific damaged path in hippocampus
- Inability to remember
    - Encoding failure
    - E.g. absentmindedness, memory decay
        - Absentmindedness
            - We can't pay attention to everything
        - Memory decay: forgetting over time
            - Forgetting curve
                - Initial forgetting very fast but then it stabilizes over time

            - Why?
                - Time (metabolic process)
                - Interference
                    - After sleep less forgetting then after wakefulness
                    - Proactive interference
                        - Old thing you studied keeps interfering with new thing (you keep calling you new partner your old partners name)
                        - The other way around: retroactive interference
- Memory reconstructions (distortion of memory)
    - E.g. bias
        - Memory bias
            - Changing of memories over time so they become consistent with current beliefs or attitudes
            - People often revise memories when out attitudes/beliefs change
        - Source misattribution:
            - Memory distribution when remembering time, place, person, or circumstance of memory/information
            - Familiarity
        - Suggestibility
            - Development of biased memories from misleading questions/information (on purpose or by accident)
            - Misinformation Effect
        - Repressed memories are controversial
        - However, if cognitive interviews are neutrally worded, the accuracy of recall increases
        - Memories can be altered by intrusion of generic knowledge

Varieties

- Explicit memory (declarative)
    - Episodic memory
        - Events in your live, personal experiences
    - Semantic memory
        - Facts and knowledge
- Implicit memory (nondeclarative) (learned skills, unconscious track)
    - Classical conditioning
    - Procedural memory

When memory fails

bed rest awake bed snooze drowsy tired

# Week 4

## Lecture 4.1 – Thinking and Language

Mental representations

- Anything to do with perception and knowledge → your mind isn't in direct contact with the environment so its represented there
- Analogical representation
    - E.g. image of cat or sound of cat
    - Dissimilar from pictures (not the exact depiction of something)
    - Network model: Nodes have relationships with other nodes → basically information cluster → associations
    - Exemplar = generalized object, like a bird
    - Prototypes = one specific object of a kind, e.g. snow owl as a kind of bird
- Symbolic representation
    - Abstract (e.g. language (written or verbal))
- Thinking = manipulation of mental representation

Decision making

- Judgement = deciding on likelihood of something when you have incomplete information
- Decision making = selecting one option among multiple possibilities
- Traditional decision research
    - Rational choice: single "correct" choice after a cognitive evaluation
    - Behavioral violarion of rational decision making
- Normative decision theory → define how people should make decisions
    - Expected utility theory:
        - Decisions should be made to maximize outcome
        - Should depend on
            - Values of alternative options
            - Possibilities associated with alternative options
- Descriptive decision theories → describe how people actually make choices
    - Heuristics:
        - Shortcuts (rule of thumb, informal guidelines etc.) used to reduce the amount of thinking that is needed to make decisions
            - Often unconsciously
            - Allows us to decide quickly
            - Might result in wrong decisions
    - Anchoring
        - Tendency, in making judgments, to rely on the first piece of information encountered or information that comes most quickly to mind (first information become reference point or anchor point; e.g. bargaining about price and first named price is anchor)
    - Framing:
        - In decision making, the tendency to emphasize the potential losses or potential wins of different alternatives
            - Value of 1000€ gain is worth less than the value of 1000€ loss
            - Risk aversion: preferring a sure gain to a risky gain
            - Loss aversion: more sensitivity to potential losses than potential gains
    - Availability heuristic:
        - Deciding based on the answer that most easily comes to mind
    - Representativeness heuristic:

        - Deciding based on how typical and individual, object or event belongs in a certain category
    - Affective forecasting heuristic:
        - Tendency for people to overestimate how events will make them feel in the future
    - The paradox of choice:
        - When too many choices are available, especially when all of them are attractive, people feel more conflict about their decision
    - Good strategy is to satisfy rather than optimize your decisions

Problem solving

- Ill-defined problems (trickier for us than well defined problems)
    - Goal directed behavior
    - Obstacles to solving problems
        - Complexity
        - Mental set/predisposition
            - Instruction/practice
            - Perceptual organization of a problem
            - Functional fixedness:
                - Objects have fixed functions and we don't think outside of those functions
    - Overcoming obstacles
        - Defining sub goals
        - Restructure the problem
        - Use an algorithm
        - Use an analogy
        - Wait for sudden insight

Language

- Language: system of communication using sounds, symbols according to grammatical structure
- Semantics = meaning
- As the brain develops, so does the ability to speak and form sentences
    - Development occurs in an orderly way (first say runned not ran as past of run)
    - Listening preferences in newborns are influenced by the language or languages spoken by mothers during pregnancy
    - Newborns can discriminate all phenomes but lose this ability after 6 months
- Human vocal tract has more capacity to make sound that any language uses
    - Moving air through vocal tract
- Aphasia:
    - Language disorder that results in deficits in language comprehension and production
    - Expressive (Broca's) aphasia: interruption of ability to speak (but can write normally etc.)
    - Fluent (Wernicke's) aphasia: interruption of understanding of language in every facet (can't understand written or spoken language, but can speak etc.)

## Lecture 4.2 - Intelligence

Introduction

- General psychology
- Differential psychology
    - Individual differences, personalities etc
- Francis Galton (1822-1911)
    - Statistical analysis of mental tests
        - Many subjects
        - Research on inheritance of specific traits to do with intelligence
        - Also wanted to prove racist things
        - Collected data on mental activities
- Pearson (1857-1936)
    - Covariation of multiple traits
        - Mathematical possibility of predicting the capacity on basis of many tests on various mental traits
- Spearman (1863-1945)
    - Intelligence: single general factor and some specific traits
- Thurstone
    - Factor analysis
- Alfred Binet
    - Assessing intellectual aptitude of schoolchildren
    - Concept of mental age
- William Stern
    - Deviding mental age by chronological age to calculate individuals intelligence quotient (IQ)
- 1916: standardized IQ test

Intelligence testing

- What is intelligence
    - Plan, solve problems, reason, think abstractly, quick learning, complex ideas etc.
    - Many different definitions
- How to measure intelligence
    - Standardized psychometric test
        - IQ = MA / CA * 100 (Binet, 1911)
            - MA = Mental age
            - CA = Chronological age
            - Works until age of 16 (afterwards IQ doesn't improve anymore)
        - Bell curve → based on standard deviation
        - Wechsler adult intelligence scale (WAIS)
            - Not necessarily culturally fair
            - Arithmetic tests, object recognition etc.
        - Many problems with tests
        - Scholastic Aptitude Test (SAT)
            - Group test
            - Not really IQ test
        - Ravens progressive metrics test

            - Nonverbal tests
        - General
            - High reliability
            - High predictive validity (depends on complexity of someone's job/task)
            - Some caveats (e.g. anchored in western society etc.)

Psychometric approach

- See if abilities are connected and have one underlying correlation
- → many of the abilities we define as intelligent are to some degree related → working memory ability/capacity (hold multiple objects at the same time) as underlying factor
- Attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests
    - Spearman
        - General intelligence (g)
            - Linguistic
            - Numerical
            - Spatial
    - Cattell: fluid (how much you can learn and change) vs. crystallized (how much you know) intelligence
        - Fluid intelligence drops with age
- Standardized intelligence tests fail at capturing other types of intelligence
    - Types of intelligence: Analytic, creative, practical, emotional intelligence etc.
- Different measures correlate with g
    - Simple RT (reaction time) and choice RT task
    - Discrimination task (is A different than B, higher g → shorter time)
    - Working memory capacity
    - Executive control (ability to switch tasks etc.)
- Score on intelligence tests is influenced by genetics and environment
    - Genetic influences
        - Twin studies
            - Identical: .86 correlation in intelligence → doesn't drop when raised apart
            - Fraternal: .6
    - Environmental influence
        - In rich environments => more variations compared to impoverished
            - Can try and fail, more opportunities etc.
- Heritability H
    - Between 0 and 1
    - H = Generic Variance / Phenotypic variance
    - 0 = everything is due to environment
    - 1 = everything due to genetics (like having 2 eyes)
    - Based on western middle class, so little reliability
    - Statistical estimate of the proportion of variance in a population
- Flynn effect:
    - Environmental changes change root score of IQ → average IQ now 107 instead of 100
        - More stimulation, better nutrition

- Now: reversed Flynn effect
- Group differences
    - Men vs women
        - Very hard to proof
        - Publication bias (want significant results)

Beyond IQ test
Relation IQ and cognitive performance
Roots of intelligence

# Week 5

## Lecture 5.1 – Scientific reasoning

General: Scientific method disconfirms

- The law of attraction
    - Book "The Secret"
    - Like attracts like → manifestation to the universe
    - Presented as law of nature that is science based
    - If you don't belief in it, it doesn't work
    - A lot of money involved
    - No scientific justification → why not?
        - Hindsight bias ("I-knew-it-all-along" phenomenon; after something happened, believing you could have predicted something)
        - Overconfidence, especially if we don't know much about a field

Fixation of belief

- Belief
    - Attitude towards particular claim
- Claim (or statement)
    - Can be true or false
- No direct connection between truth and belief
- Scientific justification (example time dialation)
- Methods to fix beliefs
    - Non-scientific methods
        - Personal experience
        - Intuition
        - Ask Authority
    - Scientific methods
        - Empirical research
            - Conclusions by combining
                - Proper reasoning
                - Systematic empiricisms
- Solvable problems
    - Science only deals with solvable problems
    - Potentially solvable
    - Observations systematic and empirical
    - Verifiable

What is science

- Empirical cycle (theory-data cycle)
    - Theory → Hypothesis (H) = statement relating theory to reality
    - Deduction → Prediction (P)
        - Top down reasoning: Conclusion about something specific, based on a rule
    - Data
    - Induction (back to theory)
        - Bottom up reasoning
        - Come up with bottom up rule that explains all of the data
- Syllogism
    - 2 premises → conclusion
    - Valid or invalid
        - Does the conclusion hold, given the premises
        - Doesn't have to be true, logic just has to hold
- Conditional argument: If A then B, Thus B valid (confirming antecedent)
    - In A then B, not B, therefore not A (denying antecedent) → don't to illicit conversion
- Hypothesis = testable prediction
    - Research = test of hypothesis with scientific methods
    - Operational definition = statement of procedures (operations) used to define research variables (e.g. how to measure aggression: frequency of behavior that aims to harm other people)
    - Initial Conditions (IC) + Auxiliary Assumption (AA) → Prediction (P) (so if IC and AA are true, P has to be true; but NOT: P is true, so IC and AA have to be true/ just because clocks run slower, doesn't mean time dilation exists)
    -
    - Research process:
        - Theory → Hypothesis → research and observation → (generate or refine) Theory
    - Replication (also in different circumstances etc.)
    -

Logic behind science

Scientific reasoning

- 2 conditions for good tests of hypothesis
    - If [H and IC and AA] then P
    - If [not-H and IC and AA] then probably not-P …. (look at slides)
- example

Falsifiability criterions

- to find good justification for the belief that a hypothesis is true, the hypothesis needs to be potentially falsifiable
    - has to be precise too

Two theories

- Evolution Theory
    - Can be falsified → scientific
- Intelligent Design Theory
    - Cannot be falsified → not scientific

## 5.2 – Validity

Basic knowledge

- Stroop task: Incongruent > neutral > congruent (time needed)
- Independent variable
    - Manipulated variables
- Dependent variable
    - Measured variable
- Confounding variables
    - Changes with independent variables but you might not be aware of it
- Control variables (constants)
    - Being held constant on purpose (otherwise would be confounding)
- Random error
    - Things we cannot control
- Conceptual variable (construct)
    - Abstract version of construct like anxiety
- Operational variable (operational definition)
    - Specific way in which a construct can be measured
- Goals
    - Describe → Predict → Explain/understand → Influence/control
- Describe: Three claims
    - Frequency claims
        - Focus on one variable
        - Descriptive method/research
    - Associative claim
        - Focus on relationship between at least to variables
            - Association
                - Correlation
                - Covariation
                - (does not equal causation (no 'let to' in claim), no directionality in claims) (correlation is not causation)
    - Causal claim
        - Focus on causation
            - Change in one variable is responsible for changing the value of another variable
            - Experimental research
                - Try to isolate causes and their effect
- Predict:
    - Mainly by way of accessing the relationship between two or more variables → don't infer causality
- How do we know whether claims are good?
    - validity
        - Reasonable
        - Accurate
        - justifiable

Construct validity

- indication of how well a conceptual variable is measured → how well we operationalize a variable
- threads:

        - inadequate operational definition
            - does it measure anxiety or absence of self-confidence?
        - mono-operation bias
            - only one aspect of a concept like anxiety?

External validity

- how well a result of a study can be generalized (to different people, contexts etc.)
- threads:
    - selection bias (participants may not be representative or comparable)
        - e.g. only using men for medical drug tests
        - (semi)random assignment to experimental and control groups important
            - minimize pre-existing differences between assigned groups
    - study setting/environment
    - specific time the study is performed is different from other times

Internal validity

- extend to which it is possible to rule out alternative explanations for causal relationship between variables
- threads
    - maturation
    - history
    - testing
    - instrumentation
    - regression to the mean
        - expected return to normal
    - attention/experimental mortality (e.g. less data on second trial then first trial)
    - selection effects
        - e.g. Simpson's paradox
            - correlation in several subsets of data, but correlation is reversed when subsets are combined
        - correlation != causation
    - design confounds
        - presence of another variable that unintentionally varies systematically with the independent variable
    - observer bias
        - single-, double-, and triple-blind studies
            - single: you analyze the date before knowing which group it was
            - double: participants also don't know what is happening
            - triple: statistics is also done by person that doesn't know the study
    - demand characteristics
        - participant guess what study is about and (consciously or unconsciously) change their behavior to please/displease you
    - placebo effects / nocebo effect

Statistical validity

- extend to which a study's conclusions are reasonable or accurate
- threads
    - violated assumptions of test statistics
    - fishing and error rate problem

        - low statistical power
            - unreliability of measures

# Week 6

## Lecture 6.2 – Correlations and experiments

Correlational research

- based on association claims
- measure not manipulate variables
- correlation coefficient (r)
    - quantify relationship
    - product-moment correlation coefficient or Pearson's r
    - measure indicated linear relationship between variables (negative or positive)
    - varies between -1 and +1
        - absolute value = strength
        - sign (+/-) = direction
    - statistical validity: does data support the claim
        - statistic validity and significance often correlate
        - outlies can affect correlation
        - is correlation statistically significant
            - not reject null hypothesis (null hypothesis = there is no difference): alpha > 0.05 (or 0.01/0.001)
            - reject null hypothesis (and accept your own): alpha < 0.05 (0.01/0.001)
    - directionality problem: we don't know what direction the correlation is (e.g. which one causes which)
    - 3rd variable problem → spurious correlation (e.g. correlation between ice cream eating and drowning)
    - Covariance: are variables correlated?
    - Temporal precedence: which came first (can't be solved with correlation)
        - Cross-legged-panel-procedure (way to use correlation for that)
            - Measure things with a lot of time apart and see if results are the same; shows time frame of what happened first
    - Internal validity (3rd variable problem) (can't be solved by correlation)
        - Way around it: Separating participants into subgroups on the basis of reasonable 3rd variable
            - If correlation in subgroups disappears with 3rd variable than 3rd variable is important
    - Sampling important
    - Correlation with r only suggest linear relationships, not, for example, curvilinear associations
    - Internal validity is low because no manipulation of variables

Basis of experimenting

- Null results (H0)
    - Independent variable did not affect dependent variable
    - Manipulation was too week
    - Insensitive measures
    - Unreliable measurements

        - Too much variability in data → bigger error bars
    - Ceiling effect
    - Floor effect
    - Not enough power (small n)

Experimental research

- Remember that randomization of groups is important
- Assigning subjects into different levels of independent variable(s)
    - Between-subject design (independent groups design)
        - Separate groups receive different levels of independent variables
        - Has control group
    - Within-subjects design (within-groups/repeated-measures design)
        - All subjects receive all levels of the independent variable(s)
        - Has control condition
        - ø

# Week 7

## Lecture 7.1 – Single-factor and factorial designs

Single-factor designs

- has a single independent variable
- often has multiple levels
- assign subjects to different levels of independent variables
    - **between-subject design**
        - if you can't undo it, it has to be between groups
        - advantage:
            - no carry-over effect between levels
        - disadvantage:
            - chance that groups might (systematically) differ (and you might not detect it) → therefore: random assignment, matching
                - random assignment
                - problem: too random, might not cancel out difference
                - pseudo random assignment
                - don't want to go too far, no hand-picking
                - matching
                - e.g. if you don't want the groups to differ in GPA: first highest GPA in group A, second highest in B, third highest in A etc…
            - use when:
                - independent variable has permanent, irreversible effect
                - risk of order effects (like practice, carryover)
    - **within-subject design/repeated measures design**
        - control conditions nor control groups
        - each subjects subjected to each level
        - advantage:
            - no chance of group difference
        - disadvantage:

- chance of order effect
    - often ruled out with pilot studies
    - counter order
        - full counterbalancing/complete counterbalanced design
            - all potential combinations
            - different order for each subject
            - you need a lot of participants → inefficient
        - Latin-square design
            - Each level exists in each time once
        - Balanced Latin-square design
            - Also each number occurs in each location, but also: neighbors are different (so balances out carry-over)
        - Random order
            - Useful with 6-10 different levels (so you don't need 6! Or 10! Different orders)
            - No participant gets same order as the others
- Use when: independent variable doesn't have permanent effect and order effects can be avoided by counter balancing and random order

Factorial designs

- Nothing to do with n!
- More than one independent variables, each level of every independent variable occurs within all levels of the other independent variable
- Numbers of cells in table = numbers of conditions (A x B = number of conditions, 2 x 3 = 6)
- 2x3 design

| | B1 | B2 | B3 |
|---|---|---|---|
| A1 | A1B1 | A1B2 | A1B3 |
| A2 | A2B1 | A2B2 | A2B3 |

- Tested variables are independent from each other and are not confounded
- Could be done with between-factor design, with within subject-design or a mixed design
- Advantages
    - Efficiency
        - Two factors can be tested within one experiment
    - Generalization
        - Compare different combinations of influences
    - Interaction between different factors observable
- Effects
    - (man effect of each independent variable and interaction of all combinations as well as three way interaction)
    - Main effect factor A
    - Main effect factor B
    - Interaction A x B

- Main effect = averaged out effect (if flat line = no effect)
- Interaction effects: effect of one independent variable depends on the level of the other independent variable
    - Calculate difference and see if its consistent → if yes = no interaction, otherwise interaction

Experimental research
- Only scientific methods that can give us (limited) causal claims
- Experimental research established internal validity
- Requires appropriate designs
    - Design confounds?

## Lecture 7.2 – Small-N and quasi-experimental designs

**Small-N studies**
- Small number of participants (mostly 5-6n, otherwise mostly lower than 20)
- Psychophysics experiments
    - Difference between individuals is very small
    - 'if you have seen one, you have seen them all'
- Behavior-change studies
    - Behavior change before and after intervention
    - Generate hypothesis instead of finding causal relation
    - Stable-baseline design (AB design)
        - A = behavior during baseline
        - B = behavior during/after treatment
    - Reversal design (ABA or ABAB design)
        - Like AB design, but then you revert back to the old behavior/treatment to test hypothesis → more certainty the change of behavior was due to treatment (but you can't unlearn skills etc.)
    - Multiple-baseline design
        - Fixes 'you can't unlearn skills' issue
        - Induction of treatment in time differs per participants (treatment started at different time, so kinda a control given, but not really)
- Case studies
    - Detailed examination of single patient
    - Tell us what can happen, not what typical happens
    - Often extreme cases
- Cant generalize
- Low internal validity
- Still important
    - Construct validity
    - external validity
    - Statistical validity

**Quasi-experimental Designs**
- Experimenter selects rather than manipulates the independent variable
    - Environmental variable (e.g. Covid-lockdown as an independent variable)
    - Subject variable (like brain damage)

- Thread to internal validity
- Pretest/posttest design (AB design)
    - O1 – T – O2 (O = Observation, T = Treatment)
- Nonequivalent control group pretest/posttest design
    - Experimental group (E): O1 – T – O2
    - Nonequivalent control group (C): O1 -  - O2
    - Matching important
- Interrupted time-series design
    - O1 O2 O3 O4 O5 T O6 O7 O8 O9 O10
    - Low internal validity, no control group
- Nonequivalent control group interrupted time series design (best quasi-experimental design)
    - E: O1 O2 O3 O4 O5 T O6 O7 O8 O9 O10
    - C: O1 O2 O3 O4 O5  O6 O7 O8 O9 O10
    - Still quasi experiment, because experimenter doesn't manipulate independent variables
- The subject variable "Age"
    - Cross-sectional study
- Internal Validity low but still aim to generate causal claim
- Still important:
    - Construct validity
    - External validity
    - Statistical validity
    -